Technical Details of D 263®

With high homogeneity, optical transmittance, and chemical resistance, as well as tight tolerances, D 263® has a formidable array of technical assets. Add low autofluorescence, a fire-polished surface, and a wide range of thicknesses, and you have a supremely versatile high-performance glass.

Outstanding and adaptable technical advantages

Wide range of thicknesses

While D 263® is available in a wide variety of thicknesses (from 0.03 mm to 1.1 mm, with the possibility of customized thicknesses on request), the geometrical tolerances of the glass remain consistently tight, with guaranteed quality for sheet, cut-to-size substrates and wafer.

High homogeneity

During the unique down-draw production process, SCHOTT closely monitors the optical properties of D 263®, resulting in highly homogenous sheets of glass with a refractive index accuracy of ± 0.0015. With an Abbe value of 55, it also meets the ISO 8255-1 standard for microscope cover glass.

Strong transmittance

The unique composition of D 263® and its top-quality blend of raw materials enables a high level of high optical transmittance in the UV to NIR range. A colorless appearance also helps to deliver optimum image quality, without unwanted colorization effects regardless of application.

Durable and reliable

SCHOTT D 263® performs consistently well in demanding conditions. Its high chemical resistance makes it particularly resilient in the consumer electronics, semiconductor and biotech fields.

Properties of D 263® M Properties of D 263® bio Properties of D 263® T eco

Material Properties of D 263® M

1. Optical properties

Optical Properties	Value
Refractive index* n_g	1.5354
Refractive index* $n_{F'}$	1.5305
Refractive index* n_F	1.5300
Refractive index* n_e	1.5255 ± 0.0015
Refractive index* n_d	1.5231
Refractive index* n_D	1.5230
Refractive index* $n_{C'}$	1.5209
Refractive index* n_C	1.5204
Abbe value ν_e	55
Photoelastic constant in (nm/cm)/Mpa	34.7

* Refractive indices: Pretreatment of samples.
Condition as supplied ["as drawn"]

2. Thermal properties

General thermal properties	Unit	Value
CTE (Coefficient of thermal expansion) α	$10^{-6}\cdot K^{-1}$ (20 °C; 300 °C)	7.2
Mean specific heat capacity c_p	in J/(g·K) (20 °C; 100 °C)	0.8
Transformation temperature T_g	in °C	557

Viscosities	Viscosity lg η in dPas	Temperature in °C
Strain point	14.5	529
Annealing point	13.0	557
Softening point	7.6	736

3. Mechanical properties

Mechanical Properties	Unit	Value
Density ρ	in g/cm³	2.51
Young's modulus E	in kN/mm²	72.9
Poisson's ratio μ		0.21
Torsion modulus G	in kN/mm²	30
Knoop hardness	HK 0.1/20	470
Vickers hardness	HV 0.2/25	510

Chemical toughening (for thickness = 0.145 mm)

Unit	Value
Temperature ϑ in °C	390
Time t in h	4
Compressive stress (CS) in MPa	290
Depth of layer (DoL) in µm	15

4. Chemical properties

Hydrolytic resistance (acc. to DIN ISO 719)	Value
Class	HGB 1
Equivalent of alkali per gram glass grains in µg/g	20

Acid resistance (acc. to DIN 12116)	Value
Class	S3
Half surface weight loss after 6 hours in mg/dm²	2.1

Alkali resistance (acc. to DIN ISO 695)	Value
Class	A 2
Surface weight loss after 3 hours in mg/dm²	88

5. Electrical properties

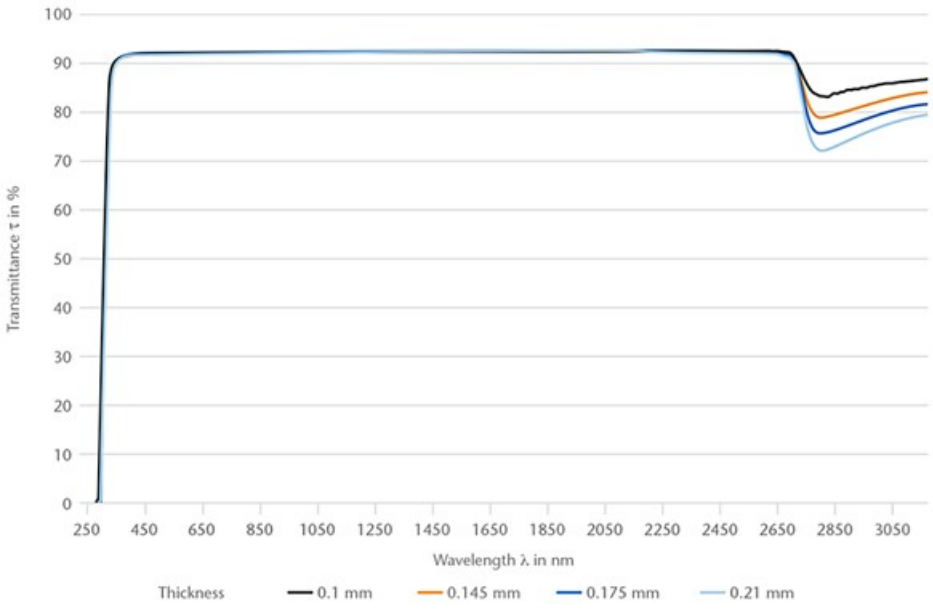
Dielectric constant εr (at ϑ = 25 °C)	Value
at 1 MHz	6.7
at 1 GHz	6.4
at 5 GHz	6.3

Dissipation factor tan δ (at ϑ = 25 °C)	Value
at 1 MHz	$61 \cdot 10^{-4}$
at 1 GHz	$74 \cdot 10^{-4}$
at 5 GHz	$101 \cdot 10^{-4}$

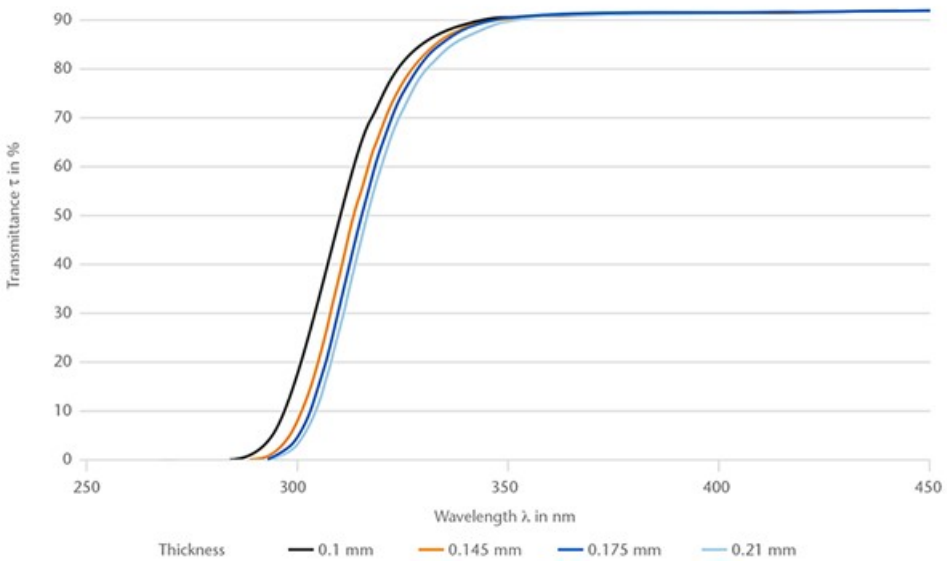
Electric volume resistivity ρD (for alternate current 50 Hz)
$1.6 \cdot 10^{8}$ (ϑ = 250 °C)
$3.5 \cdot 10^{6}$ (ϑ = 350 °C)

6. Transmittance values

Spectral transmittance: 250 - 3150 nm



Spectral transmittance: 250 - 450 nm



Wavelength in nm at thickness 0.145 mm:	τ (λ) in %
254	< 0.1
380	91.2
632.8	91.9
1064	92.2

Edge wavelength λc (τ = 0.46) at thickness in mm	Wavelength in nm
0.100	308
0.145	312
0.175	314
0.210	315

Luminous transmittance at thickness in mm	τ_{vD65} in %
0.145	91.7 ± 0.3

Want to know more? Let's talk

Whether you need more information, samples, a quote, or advice for a project, we would be delighted to talk to you.

Contact us

